Data analysis and bio-statistics

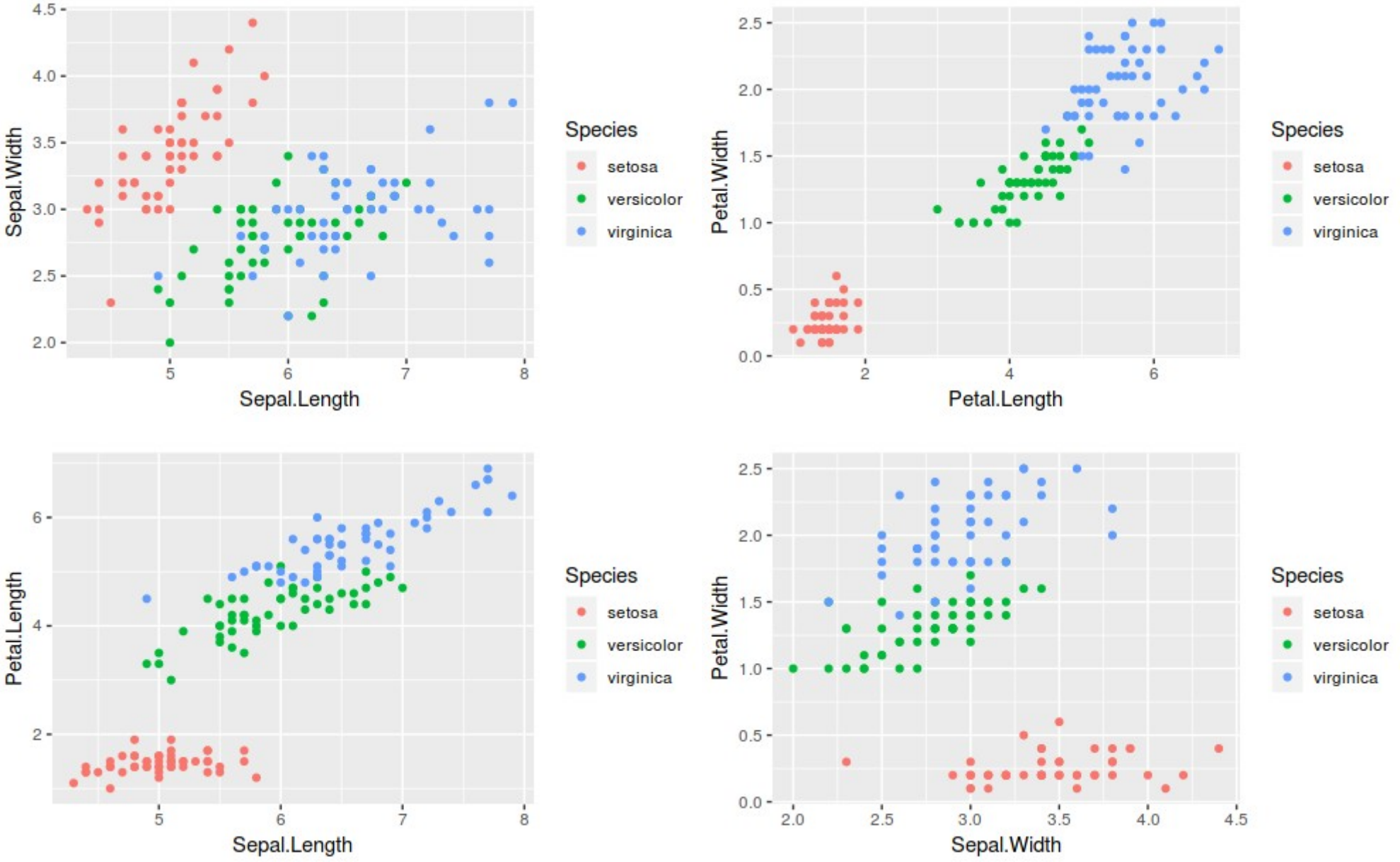
Practice 6:

Introduction to multivariate statisitcs: PCA & k-means

UNIVERSITAT ROVIRA I VIRGILI

PCA: Introduction

We can look at the data using scatter plots…



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Well… we'll have to take a look to all possible combinations.

For example, the Iris dataset contains four variables:

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- Petal.Length
- Petal.Width

So, we'll combine them all in scatter plots:

- Sepal.Length vs. Sepal.Width
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Maybe we can do a 3D scatter plot...

PCA: Introduction

We can look at the data using **3D** scatter plots…

So, we'll combine them all in 3D scatter plots:

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- Sepal.Length vs. Sepal.Width vs. Petal.Width

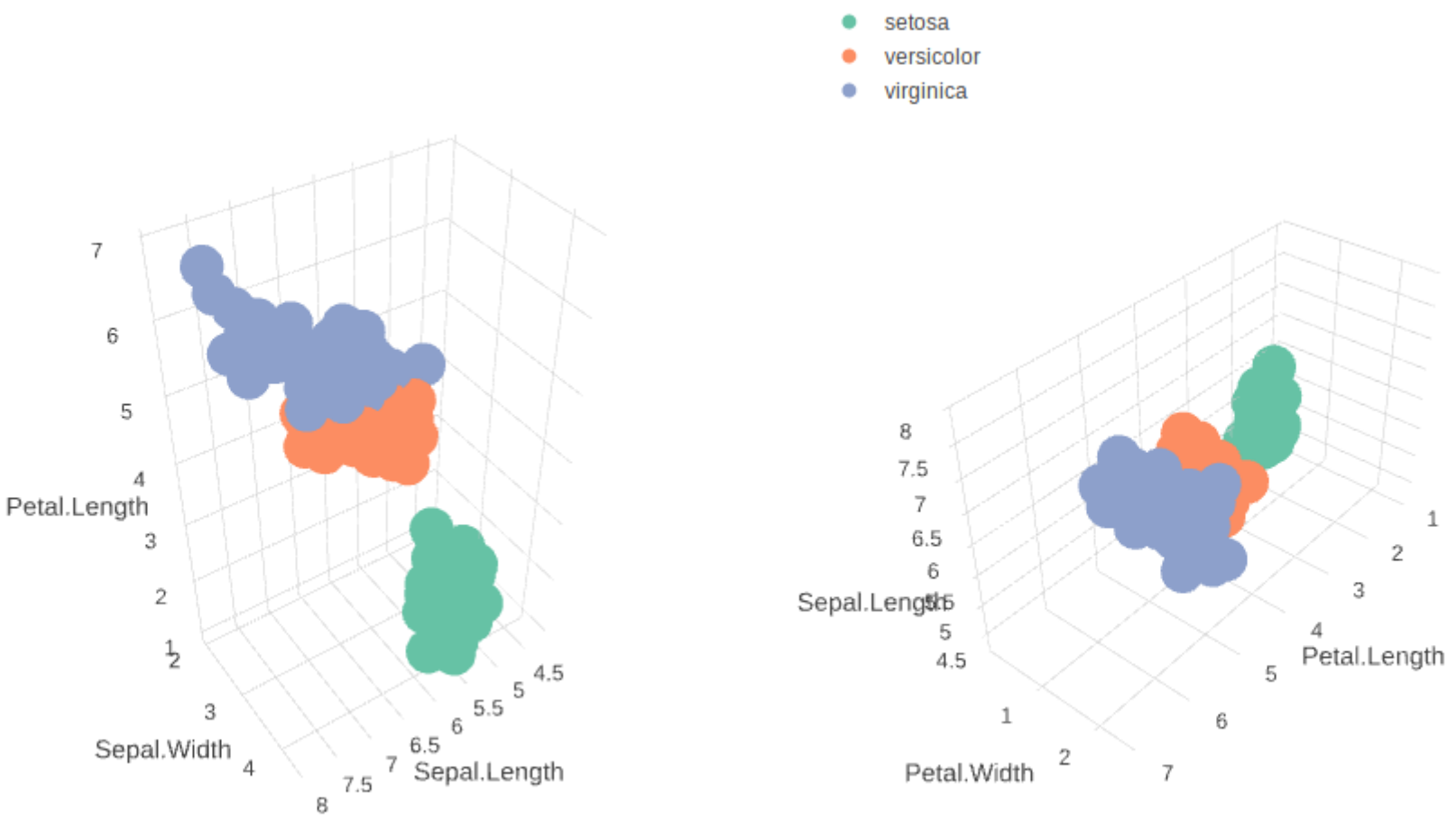
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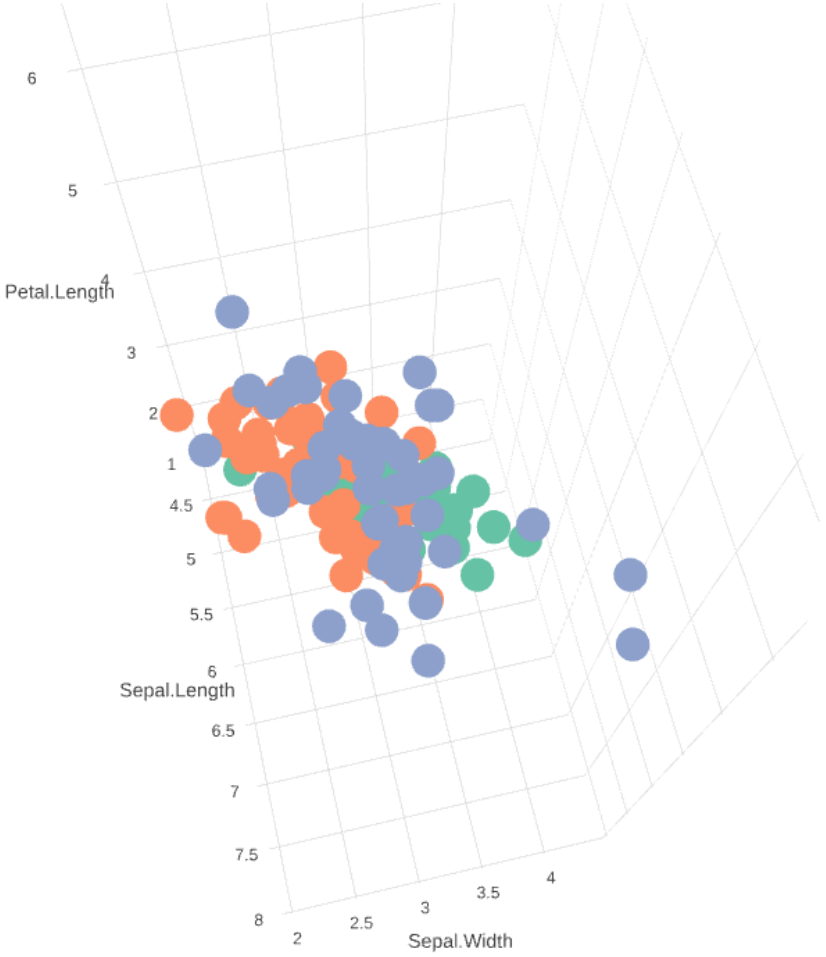
PCA: Introduction

We could look at the data using **3D** scatter plots…

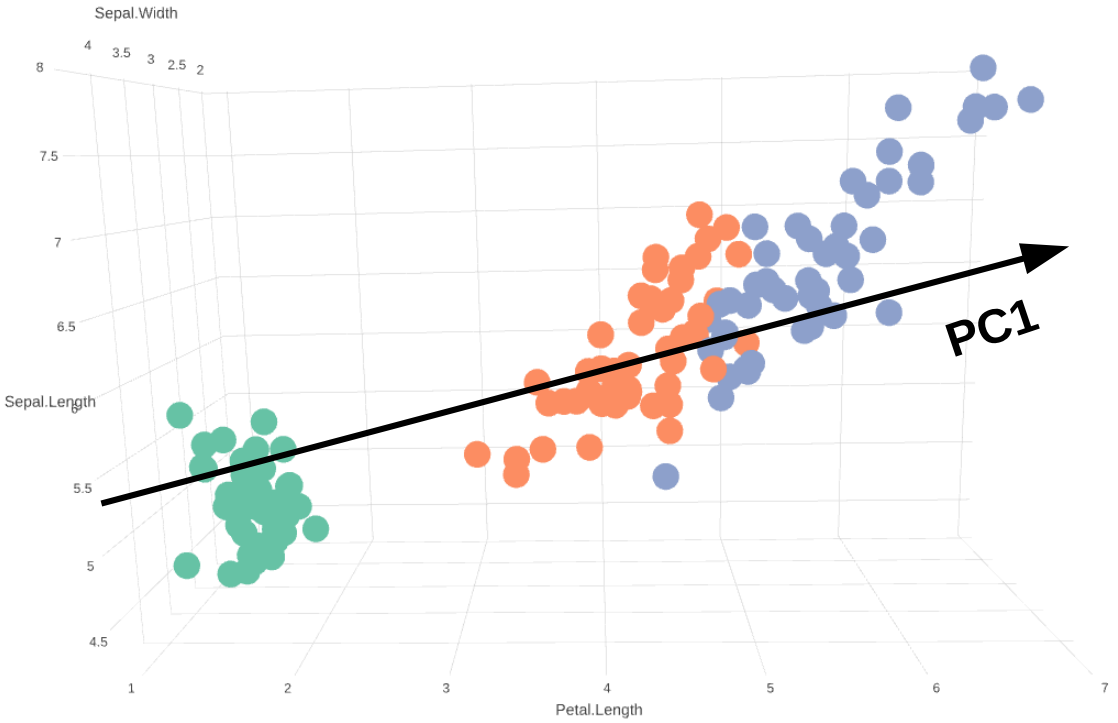


… but this is still a mess and there are only 4 variables in the iris dataset!

PCA: Introduction

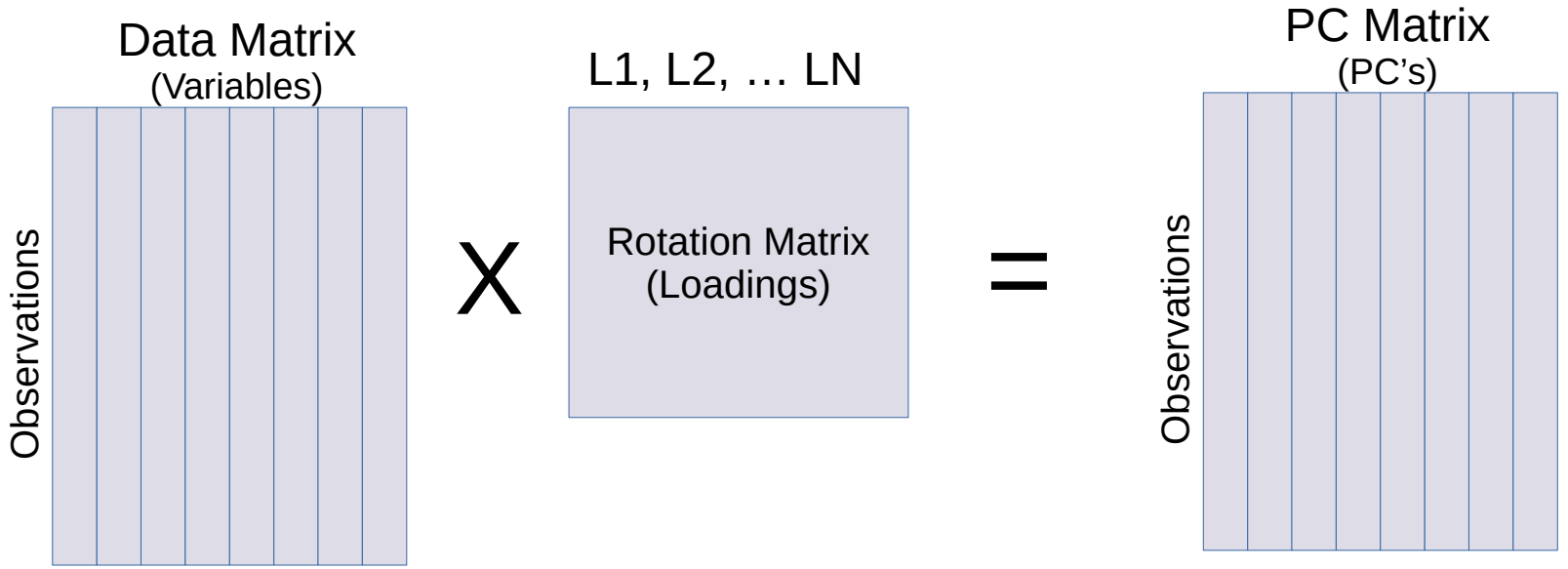


PCA Rotation



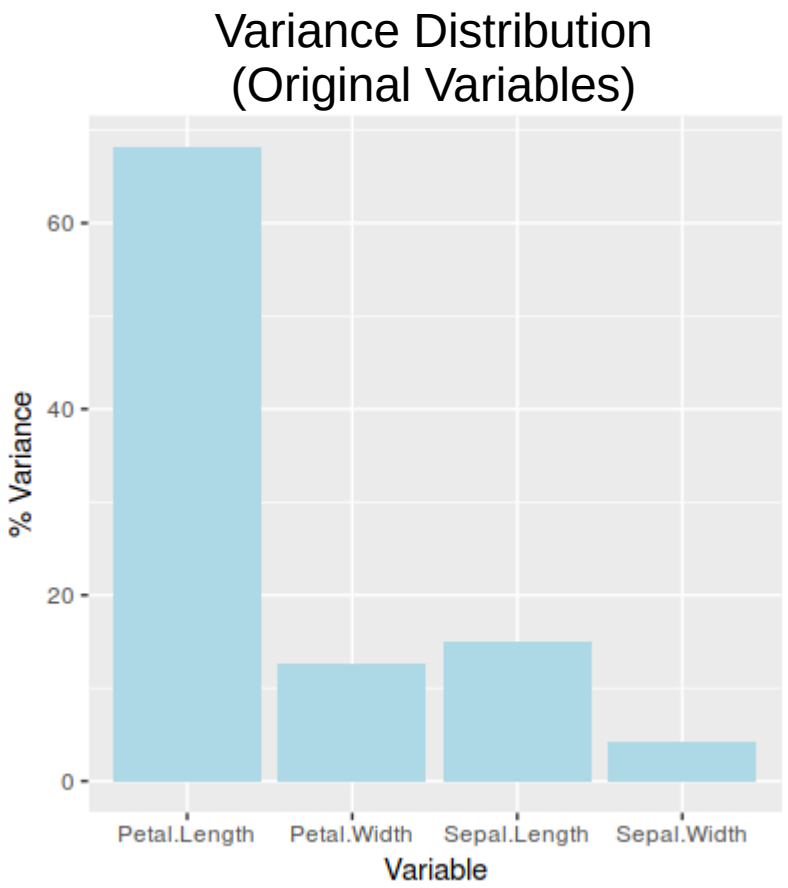
PCA: Introduction

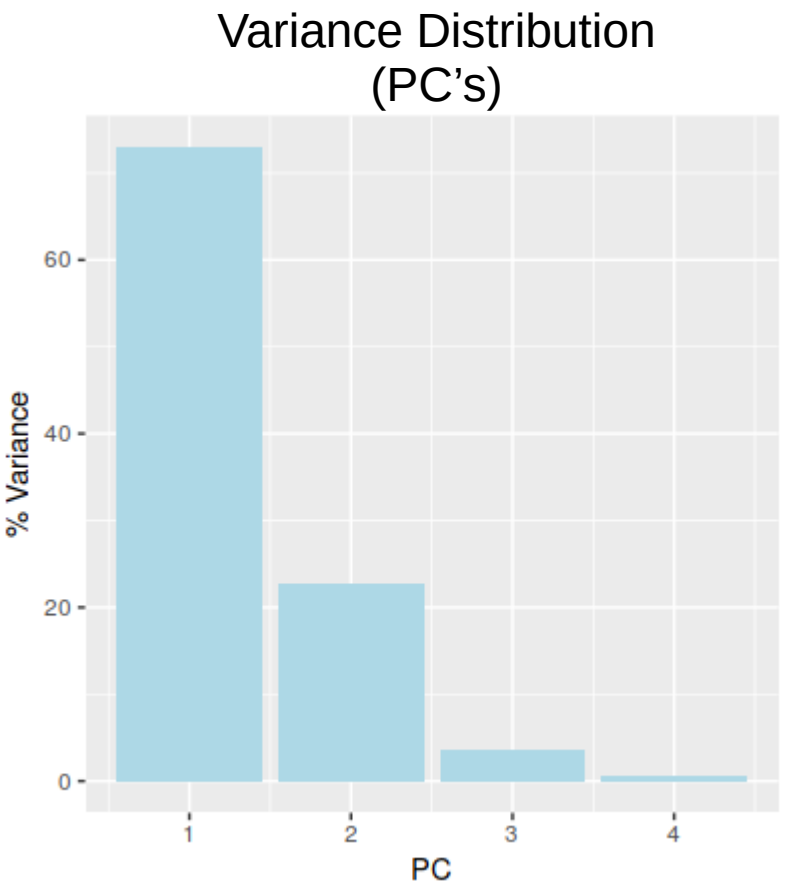
PCA transforms the original variables into new variables named principal components (PC)



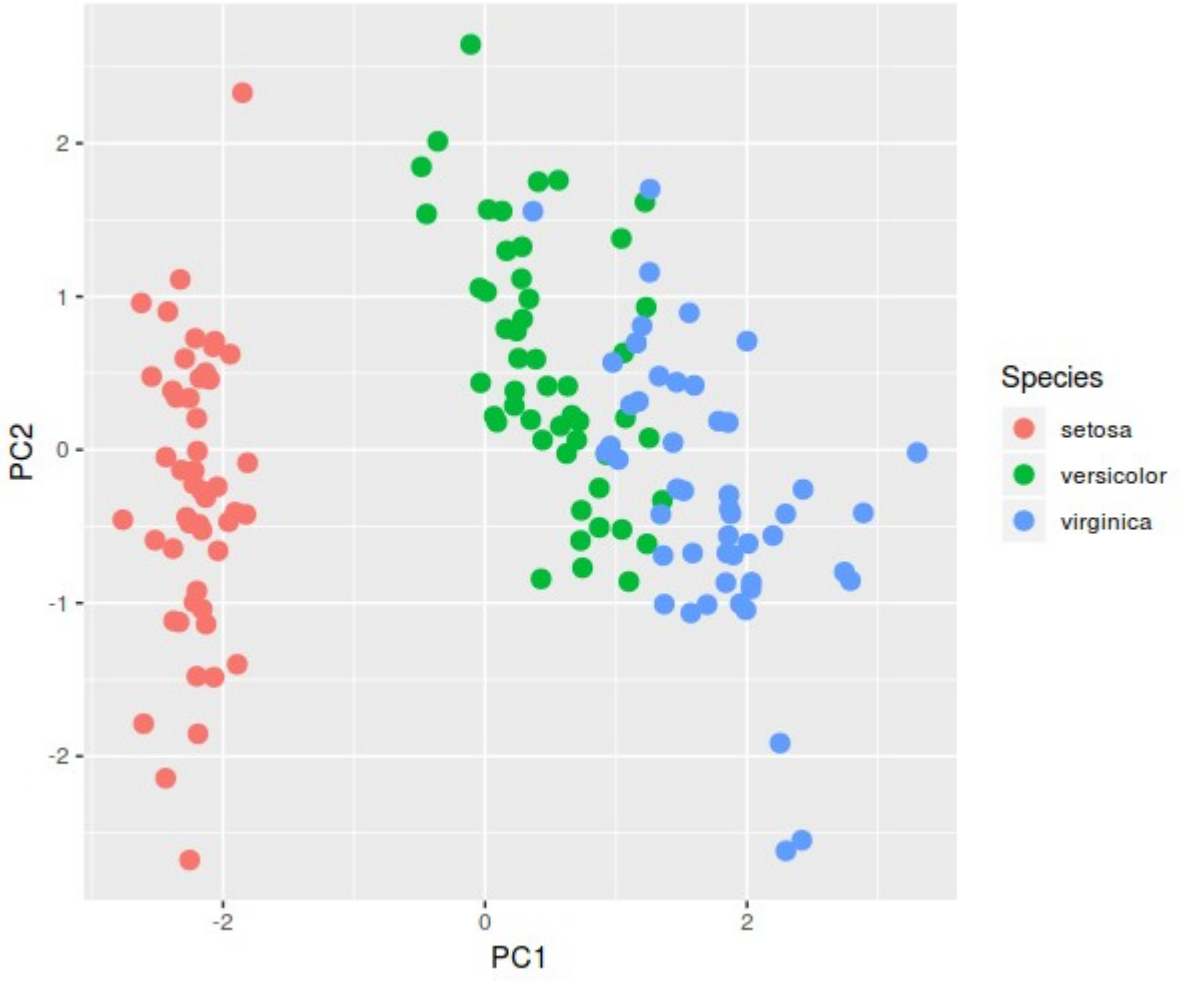
$$PC\,1_1 = L\,1_1 \cdot Var\,1_1 + L\,1_2 \cdot Var\,2_1 + L\,1_3 \cdot Var\,3_1 + \ldots + L\,1_N \cdot VarN_1$$

PCA: Dimensionality reduction

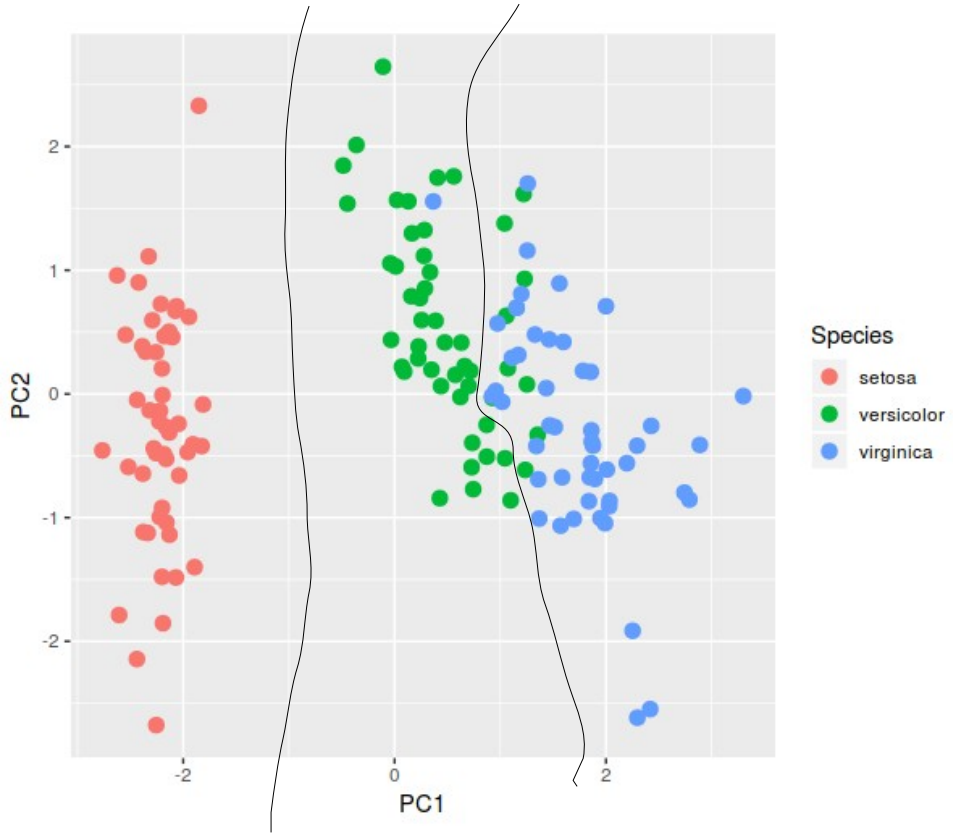


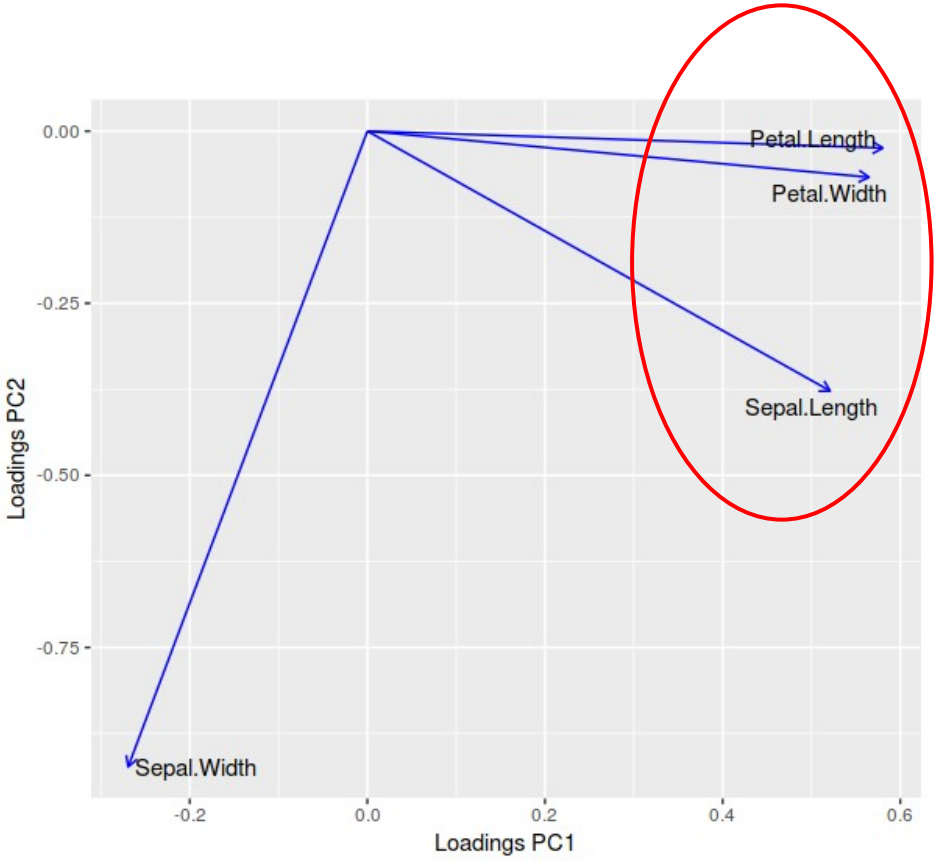


PC1 vs. PC2 (Iris dataset)

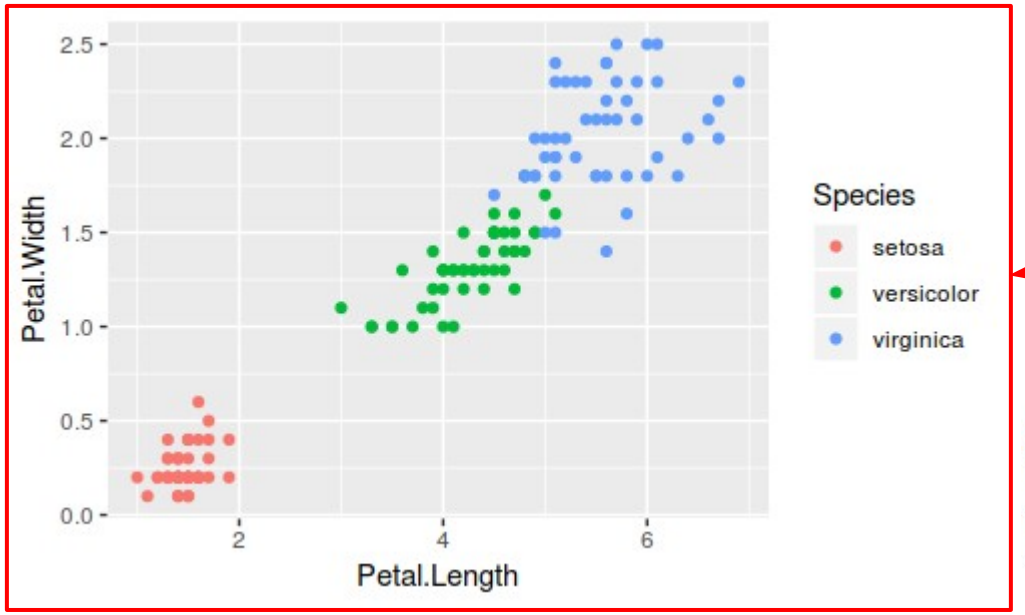


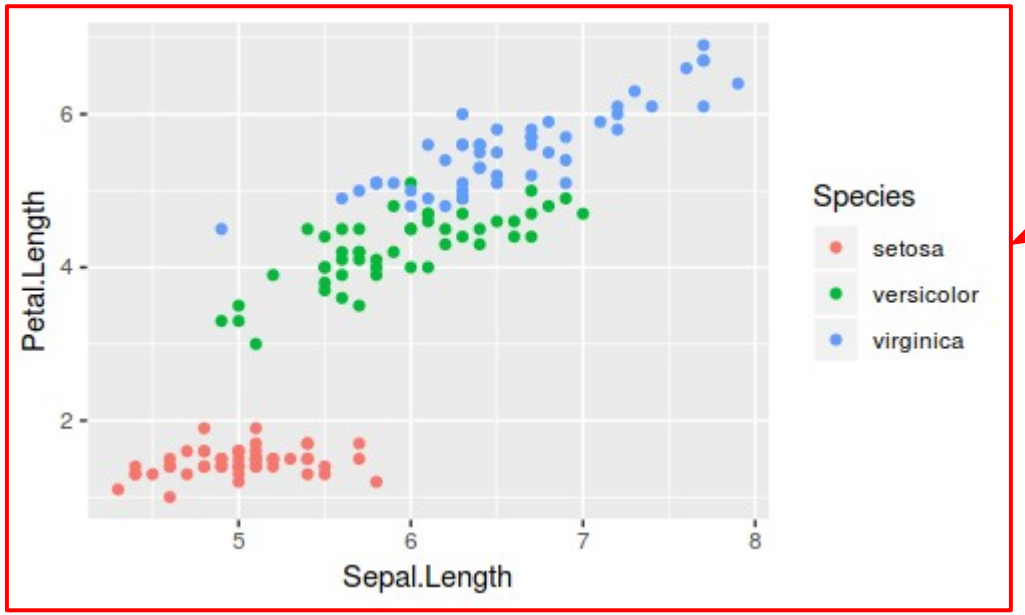
Loadings Analysis

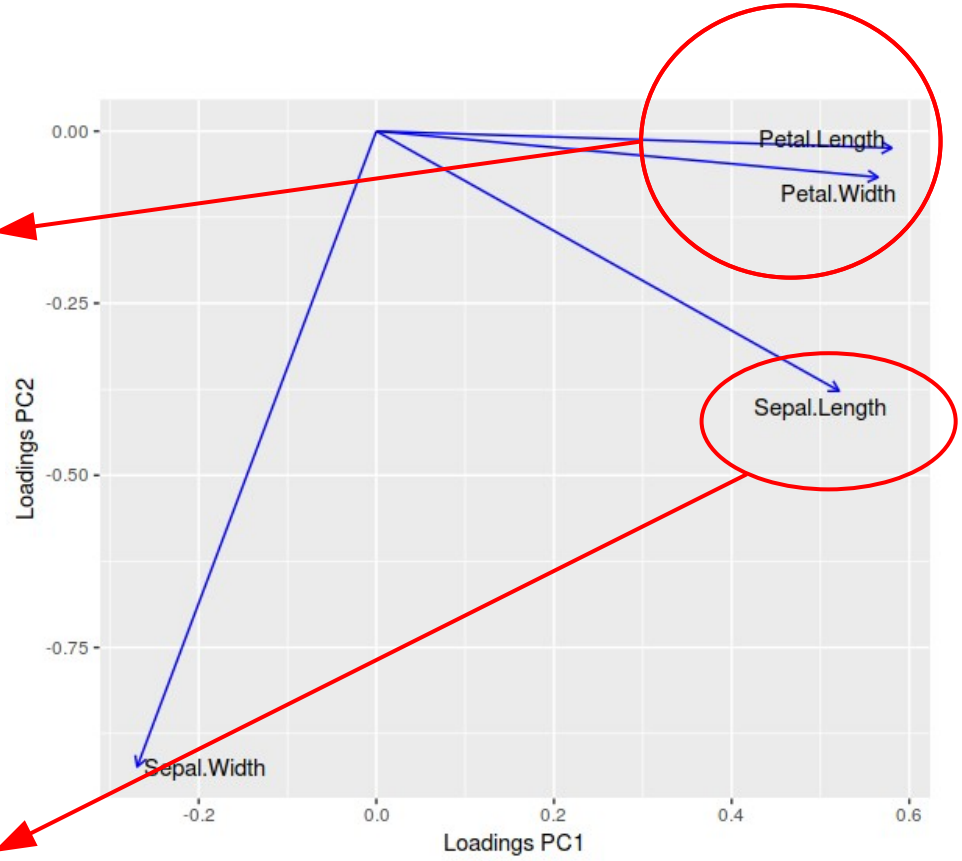




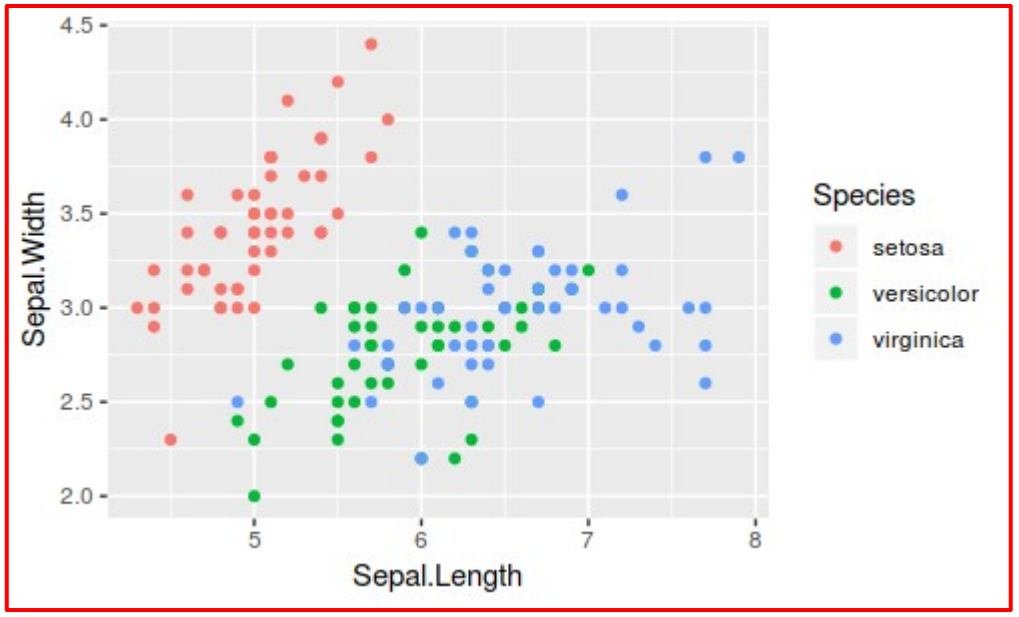
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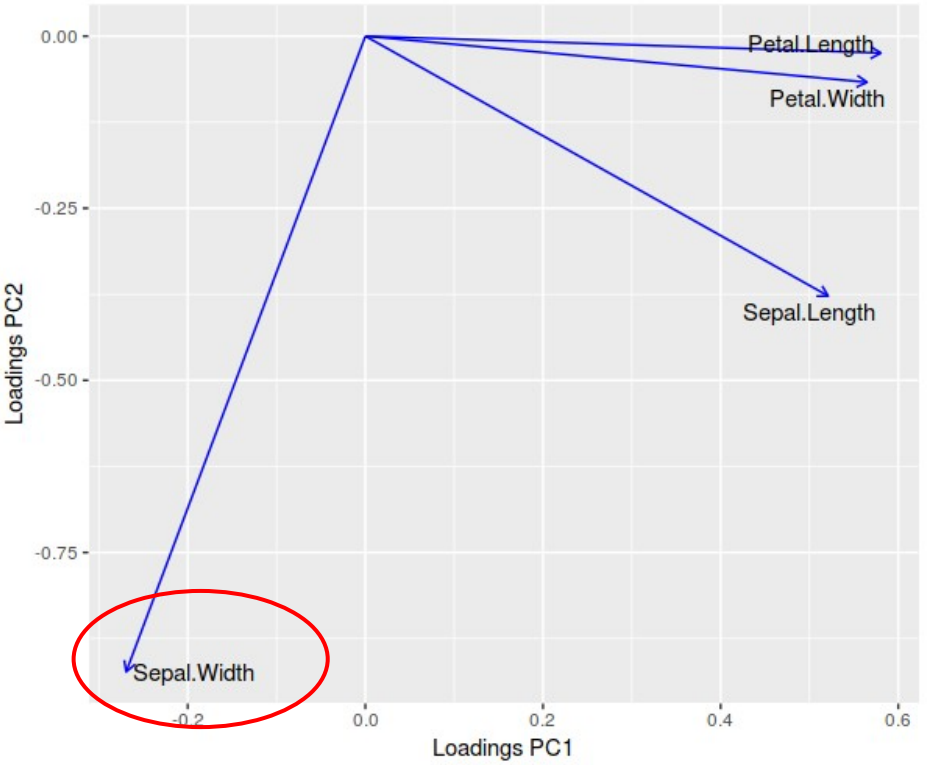




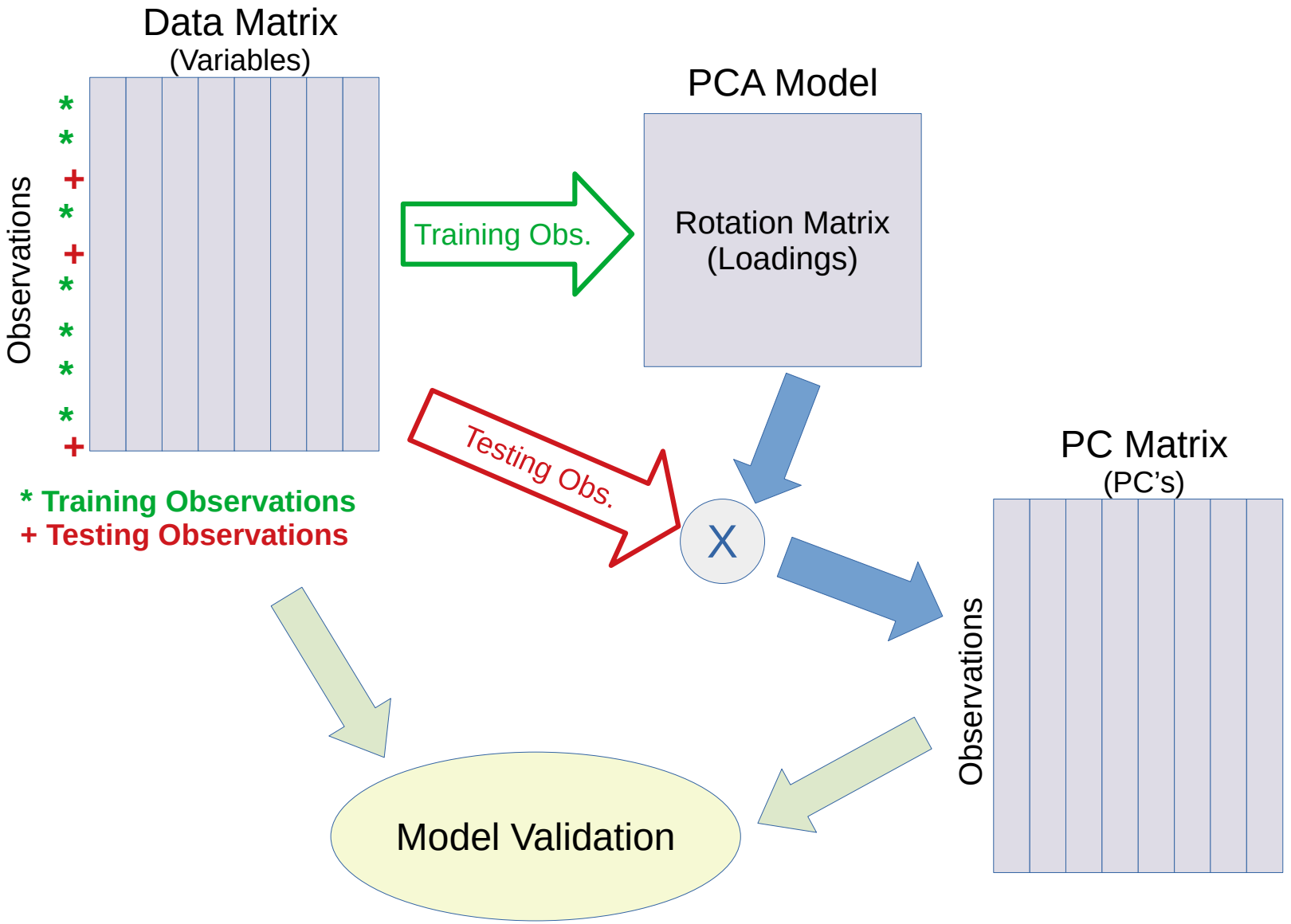


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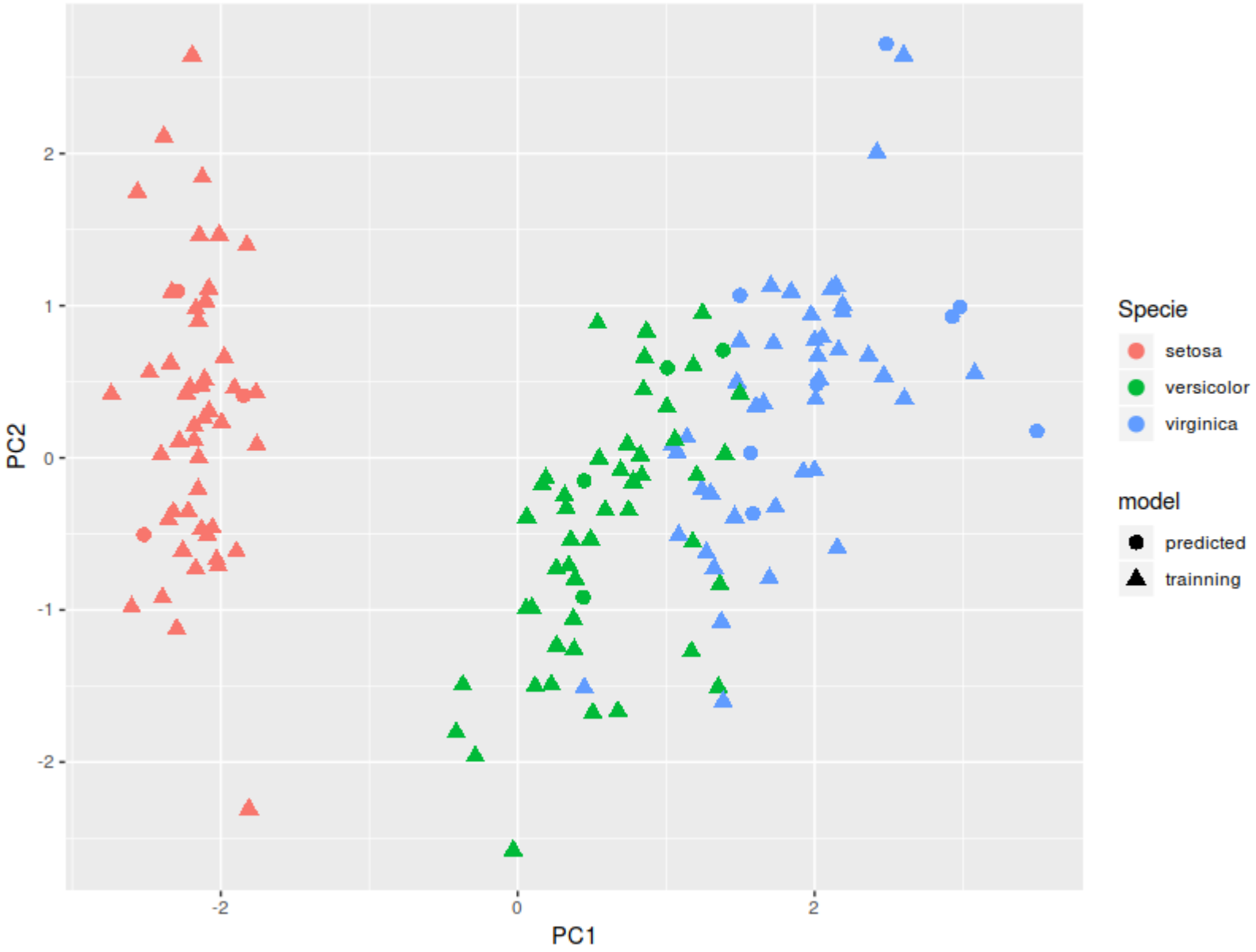




PCA as a machine learning algorithm



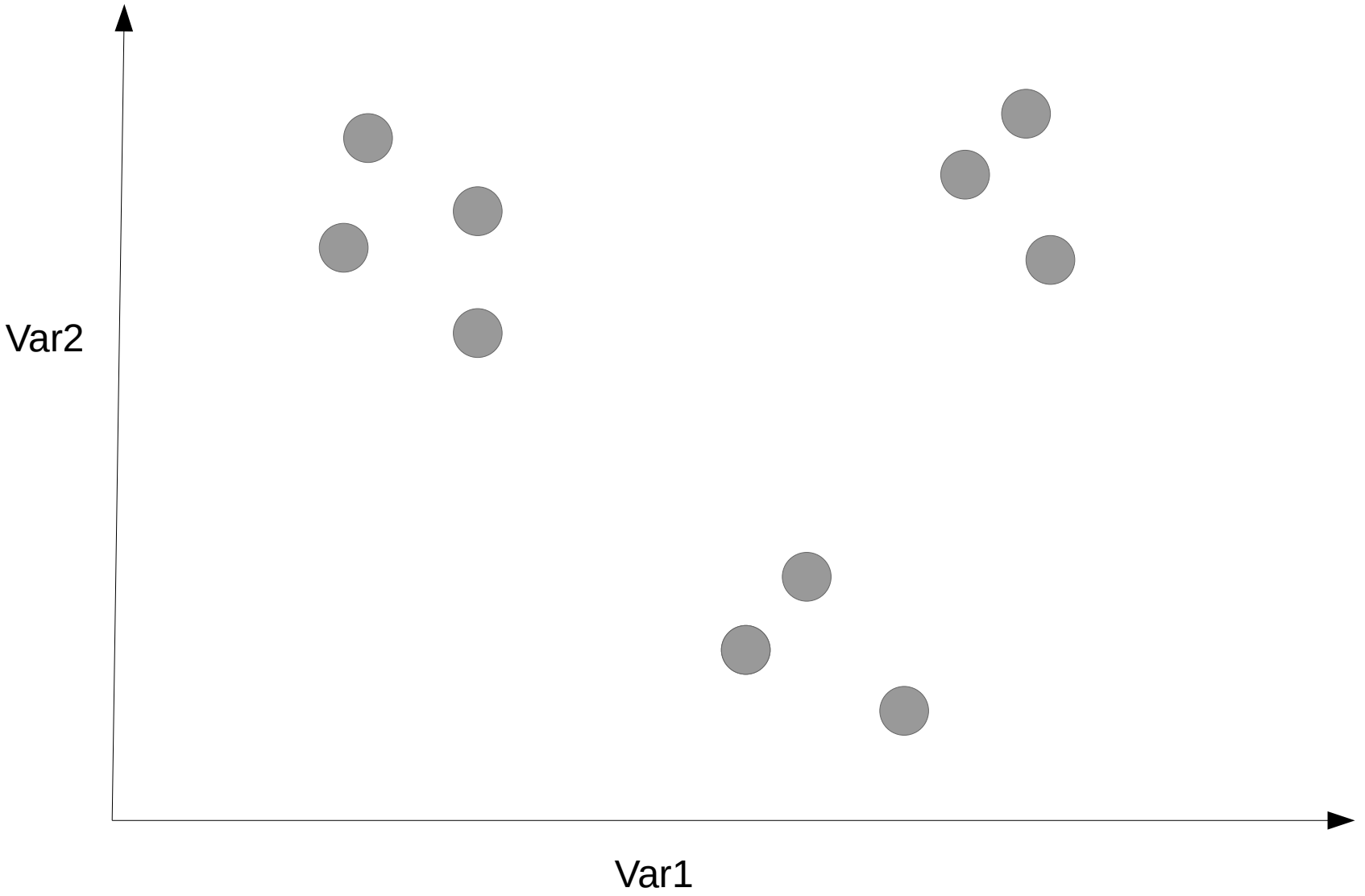
PCA as a machine learning algorithm



k-means

- Clustering? → Automatic classifier
- Unsupervised learning algorithm that tries to cluster data based on their similarity.
- Unsupervised learning: there is no outcome to be predicted (the algorithm just tries to find patterns in the data)
- In k-means clustering: we have to specify the number of clusters we want the data to be grouped into.

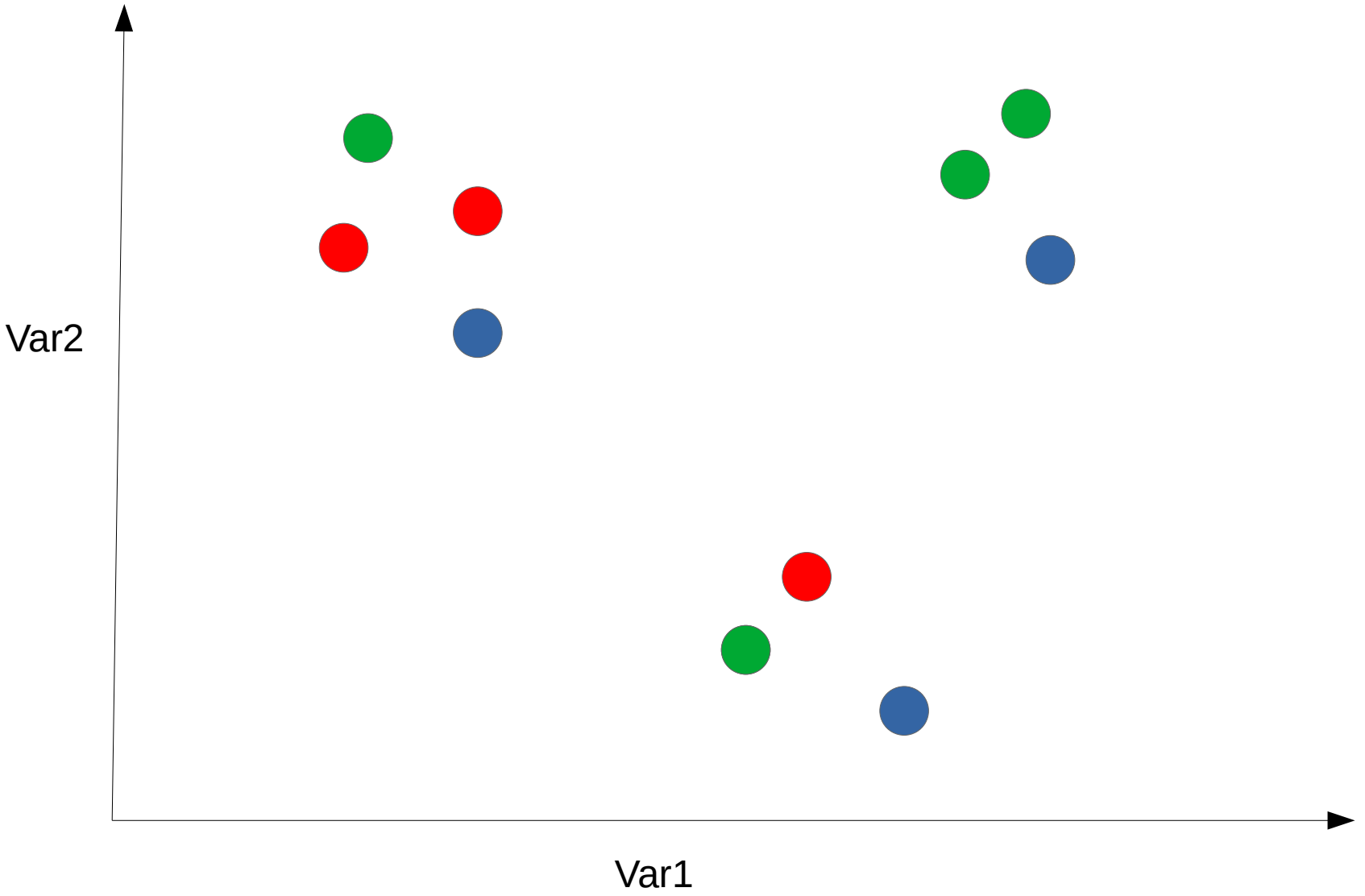
K-means: RAW data



K-means: Step1

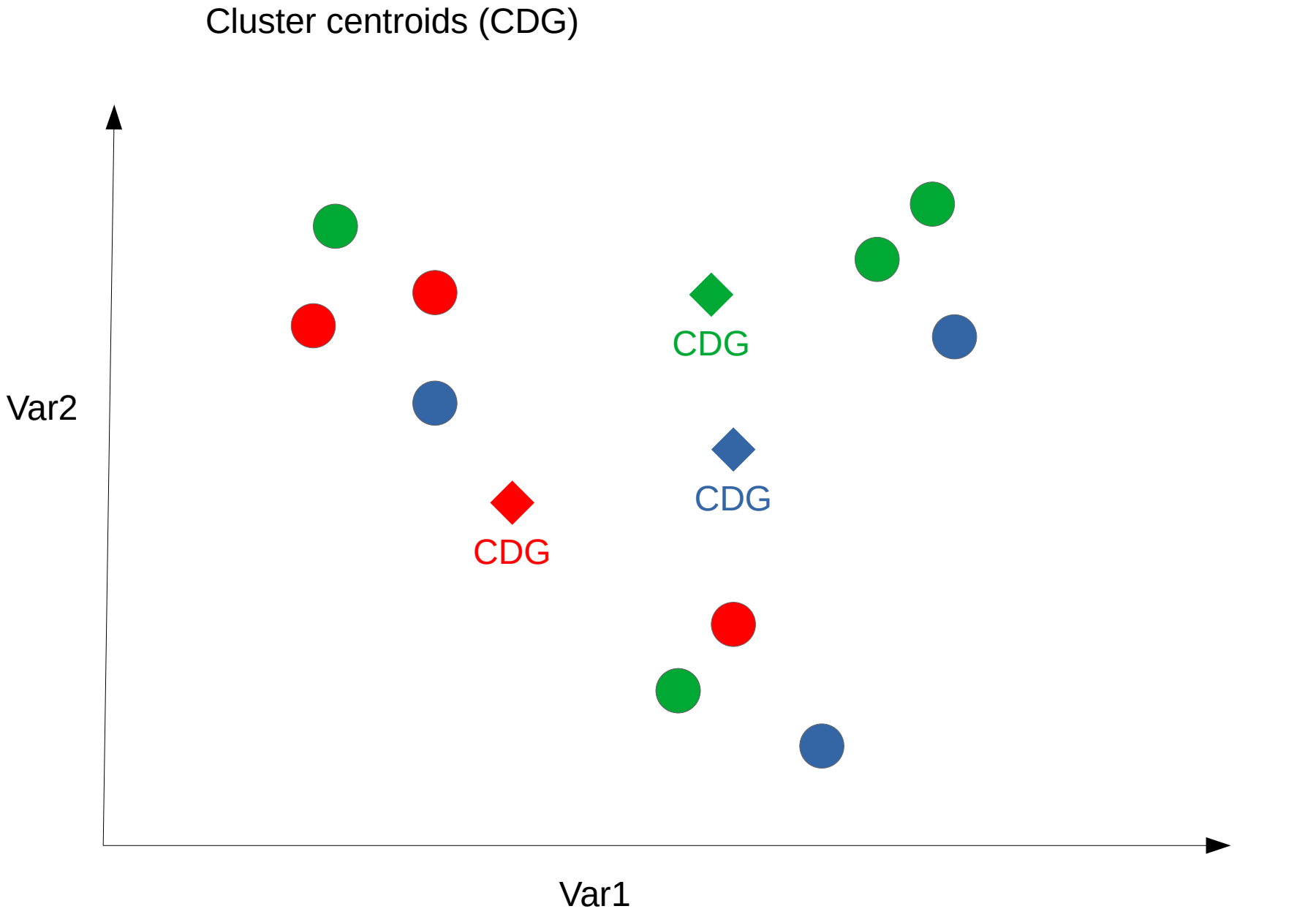
→ 3 Clusters

Random cluster assignment



K-means: Step1

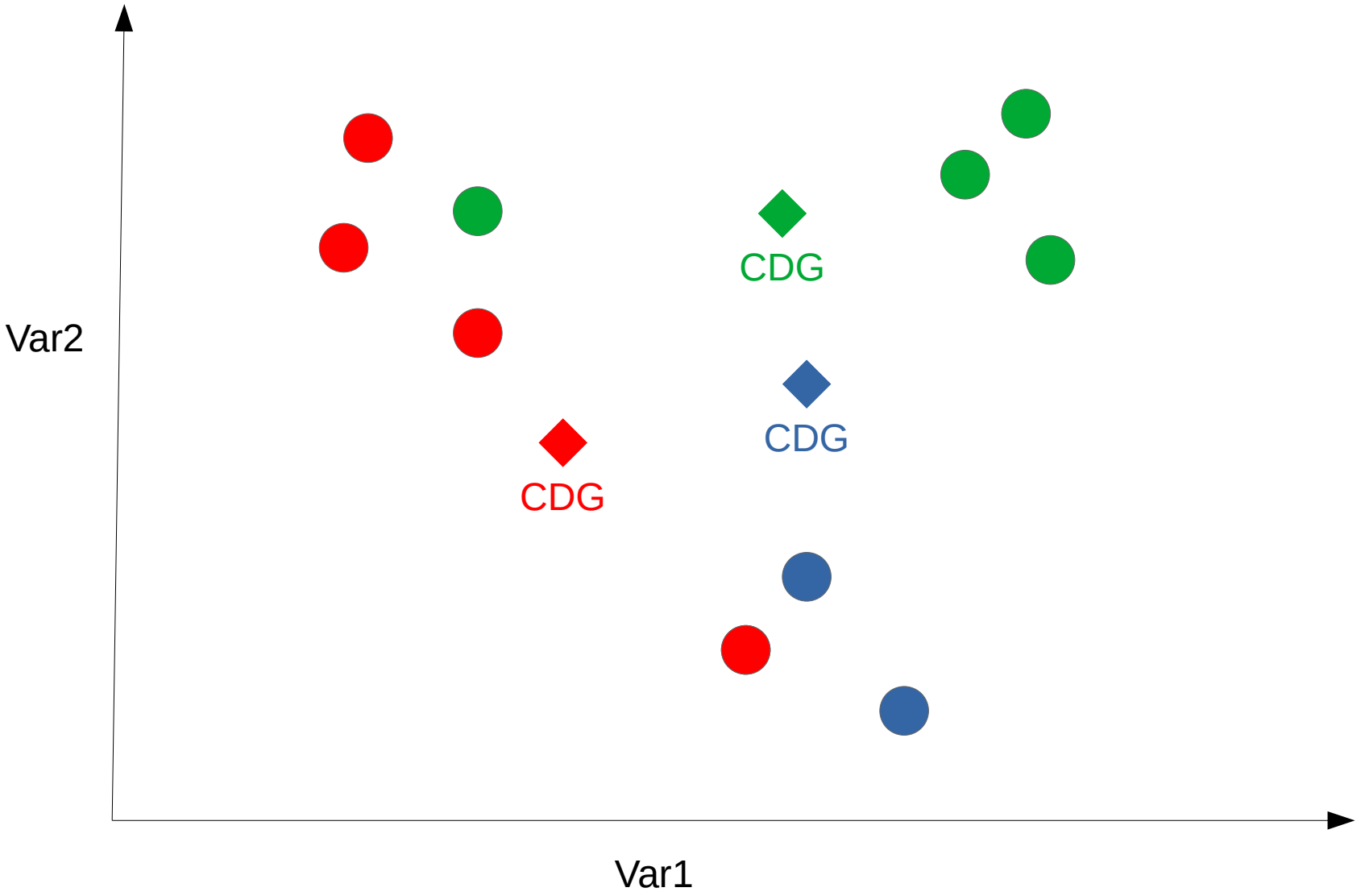
→ 3 Clusters



K-means: Step2

→ 3 Clusters

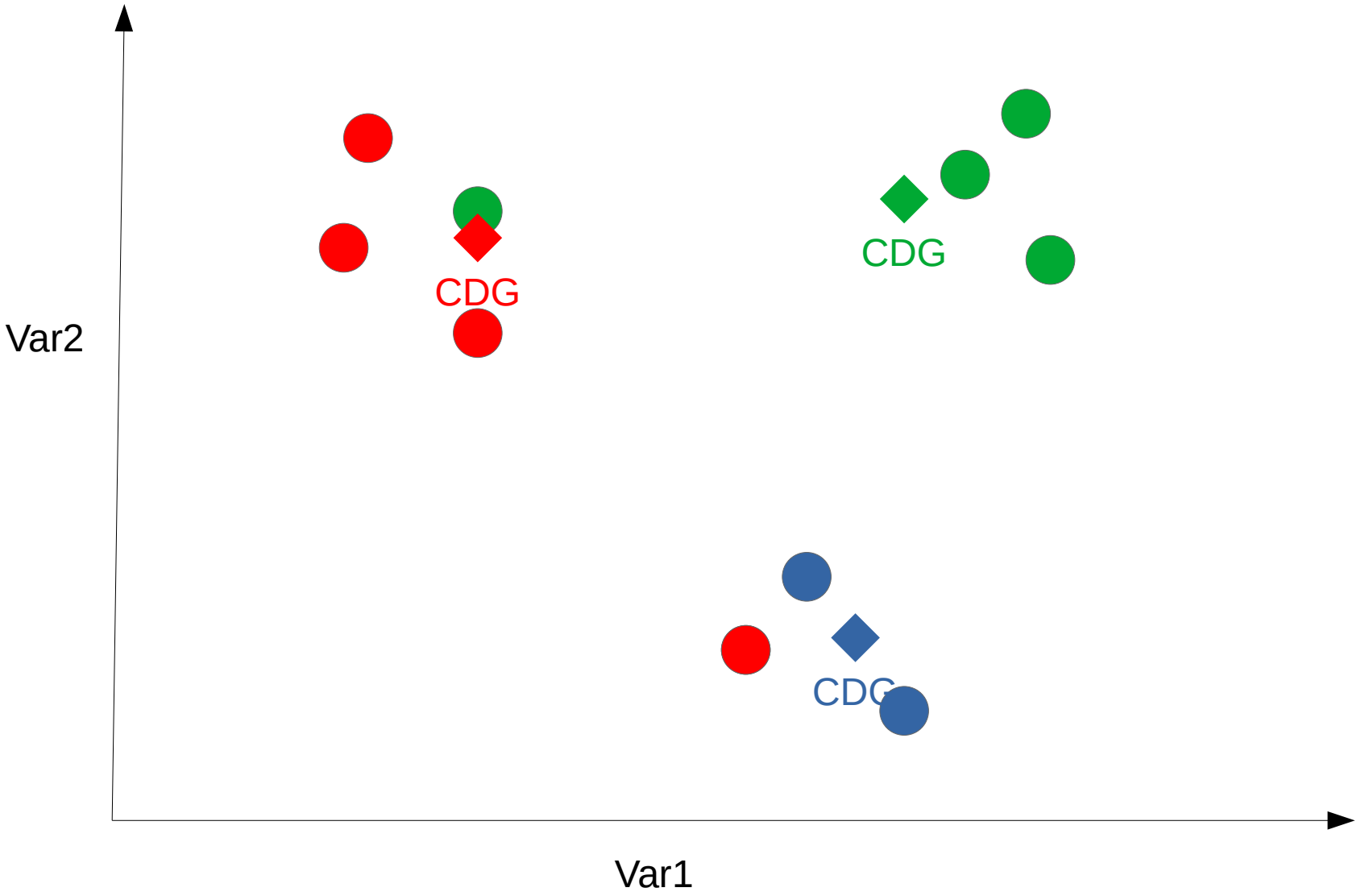
Cluster reassignment to nearest CDG



K-means: Step3

→ 3 Clusters

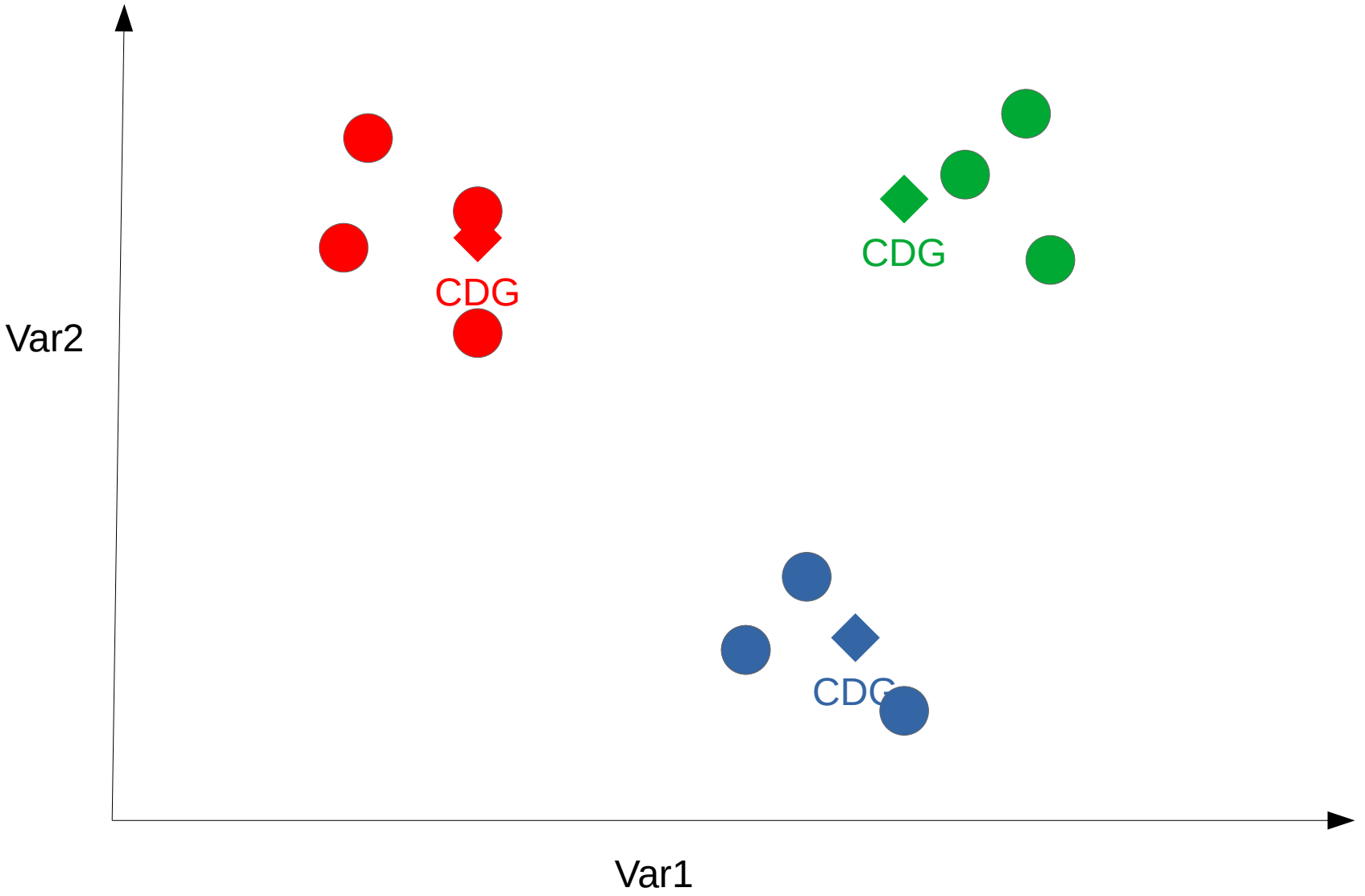
Recalculate CDG's



K-means: Step4

→ 3 Clusters

Cluster reassignment to nearest CDG



K-means: Step 5,6,7...N

Repeated till the within cluster variation cannot be reduced any further…

or… we hit the iterations limit

K-means: Result

